

MICA: Measuring Whether an AI Can Understand a Builder’s Situation and Intentions from Raw Observation

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Abstract

Can an AI watching a person build recognize what is being made and anticipate what comes next — from raw observation alone, with no instructions and no privileged game state? I built and instrumented a full pipeline for this question in Minecraft: frozen 2D foundation-model features (VPT, MineCLIP) and frozen 3D shape embeddings (Uni3D) feed a Bayesian belief over building goals, which gates an embodied assistant. The claim of this thesis is not that the system “understands”; it is that I built an honest measurement apparatus for that question and report what it measured. Five results stand: intent inference is *necessary* once builds leave templates (a structure-only baseline collapses to 0.042 final-goal accuracy on 24 never-seen real sessions while the belief tracker holds 0.375, chance 0.2); the system anticipates a builder’s next actions far above chance (71% next-action, 59% eight-actions-ahead token accuracy vs. 1.2% chance, on held-out real sessions); structural understanding accumulates lawfully with observation (partial-view embeddings converge to 0.92 cosine with the full structure); the system improves lawfully with the right data (two pre-registered experiments, both won); and the apparatus itself is trustworthy — its falsification criteria fired, a feature leak was found and a published verdict honestly reversed, and every negative is banked. Equally important are the claims I do *not* make: early recognition is statistically unproven, confidence is uncalibrated, the belief adds no measurable prediction fuel, and the agent has never placed a block — assistance is designed and safety-gated, never evaluated.

1 Introduction: why this question

In structured work environments — construction sites, warehouses, assembly lines — an assistant that must be *told* everything is bottlenecked by the telling. The expensive resource is the human’s attention, not the robot’s labor. Cooperative inverse reinforcement learning formalized this a decade ago [1]: assistance without explicit instruction is a game in which the helper must infer the human’s hidden objective, and the belief over that objective is the object everything else depends on. Benchmarks built on that frame established both directions of the necessity argument: goal-agnostic helping fails, and a confidently wrong helper is *worse than none* — undoing the human’s work in 40% of episodes in Watch-And-Help’s random-goal condition [2, 3].

But that literature runs on oracle state: its assistants read ground-truth symbolic scene graphs and, usually, labeled human actions. Robots do not get oracle state. This project asks the assistance question under the conditions robots actually face — perception in, calibration measured, authority earned — using Minecraft building as the testbed. Building has the three properties that make results transferable: discrete observable actions, goals legible from partial

structure, and cheap dense observation of a shared workspace; and open-ended building resists programmatic goal specification [6], which is exactly what makes it a genuine test of helping without being told. Domains with the same three properties — bricklaying, timber framing, warehouse picking, lab bench work — are in scope for the approach; domains without them are honestly out.

Two research questions structure everything: **(RQ1)** can the system recognize the *situation* — what is being built — from 2D and 3D perception alone; and **(RQ2)** can it understand *what the player is doing* — track the intention behind the actions — and anticipate what comes next?

2 Related work: what was inherited, what is new

2D perception. Every Minecraft foundation model I use was built to make an agent *act*: VPT as a behavioral prior cloned from ~70,000 hours of human play [7], MineCLIP as a learned reward [8], STEVE-1 as an instruction follower [9]. MICA runs the same models frozen and inference-only, pointed at a human: VPT’s trunk becomes a behavior embedding (h_{2d}), MineCLIP’s reward path becomes per-goal evidence (s_{goal} , video-text similarity against the five goal prompts). STEVE-1 is the pivot: it *writes into* the MineCLIP goal space to generate behavior; MICA *reads from* that space to infer which goal a human’s behavior is pursuing. CLIP4MC’s progress-sensitive similarity [10] is the recorded upgrade path, held behind an A/B gate.

3D structure. From SpatialLM [11] I take the representational lesson — a scene serves downstream reasoning as structured, inspectable elements — while rejecting learned scene parsing for v1: structure features are deterministic template matching (completion, edit distance, registered footprint fit) over native voxels, because converting lossless voxel state into synthetic point clouds would manufacture a backward domain gap [12]. Where 3D world models must reconstruct geometry from partial sensing, MICA’s event-sourced state makes reconstruction unnecessary; embodied partial observation re-enters only through the agent’s own raycast scan, where it is measured, not assumed (§4.3).

Belief tracking. The tracker $b(goal, mode)$ joins two lineages. From POMDP-style shared autonomy: the decide-from-an-explicit-belief principle [13] and the recursive Bayes filter over goals from non-verbal cues [14], whose score-against-pre-action-state convention became MICA’s snapshot rule. From Bayesian inverse planning: SIPS’s insight that a latent internal process between goal and action keeps posteriors sane under suboptimal behavior [15] (the ancestor of MICA’s mode axis z); CLIPS’s rule that the assistant’s own actions are interventions, never evidence [16] (MICA’s assumption A7, the do-operator); and GRAIL’s learn-per-goal-likelihoods-from-data route [17]. What none of them did: run the explicit filter *online on real free-form building*, with likelihoods *learned from frozen perceptual features* — no planner, no hand-set probability tables, no oracle state — and then measure calibration, the quantity this lineage never tested [19].

Assistance. AssistanceZero [4] is the closest prior art — the same Minecraft building-assistance game with hidden goals — but its goal uncertainty is an amortized head inside an MCTS planner over symbolic ground-truth voxels. Across the nearest siblings [4, 2, 3], the assistant receives oracle state; none infers from perception, none audits its belief’s calibration, and none asks the

human before acting. Carroll et al.’s warning that partner models not grounded in real human behavior collapse on contact with real humans [5] is why MICA’s likelihoods train on real sessions and why the builder’s own label is treated as ground truth.

Action generation. The decoder takes its architecture from multi-token prediction [20] and its recipe from Fast SceneScript [21] (NTP warm-start, then an MTP retrofit with agreement-trained per-token confidence), adopted with the source papers’ negative results conceded in writing. What that lineage did not anticipate: conditioning generation on an inferred belief about *another agent’s* intent, with strictly one-way information flow — decoder confidences never become evidence, because both source papers show those scores are not probabilities.

3 System architecture

Figure 1 shows the pipeline as currently built (the diagram is the updated version of the project’s design figure; every difference from the original design is itself a result of this project’s audits).

A Fabric capture mod records proof-grade sessions (every block event with actor attribution, POV frames, input state). Three design properties matter for everything downstream. *First, the belief is the only moving part:* the four experiment arms swap the intent module alone — reactive, implicit classifier, LLM reader, belief tracker — so any performance difference is attributable to the intent mechanism, not the harness. *Second, evidence is causally clean:* the snapshot rule (evidence strictly precedes the action it scores), the do-operator (the agent’s own blocks are interventions, never evidence), and the one-way constraints in Fig. 1 make the recorded traces intervention-grade — the leak audit that reversed the contested-labels experiment (§5) is this discipline earning its keep. *Third, authority is graded and earned:* the FSM’s lattice moves downward instantly and upward only through hysteresis, and the terminal placement state is reachable only through calibrated confidence (currently, honestly, unreachable) or the human’s per-block consent.

3.1 Compared with intention-recognition game systems

MICA’s nearest relatives are the assistance-game systems [4, 2, 3] and classical goal-recognition frameworks [15, 16]. The comparison is honest in both directions (Table 1).

The trade is deliberate: game systems optimize assistance quality inside a simulator’s epistemically generous regime; this project optimizes for the question “what would it take to trust this outside one.”

4 Results: five claims, each with its data

4.1 Claim 1 — Intent inference is necessary off-template

Data. Four-arm comparison over 24 real free-build sessions that no arm ever trained on (exposure-verified splits), final goal accuracy against 0.2 chance: reactive structure-only 0.042; implicit classifier 0.125; LLM reader 0.292; belief tracker over v0 heads 0.250; belief tracker over v1 heads **0.375** (Fig. 2).

Reading. The reactive baseline is the strongest helper that skips intent inference — exactly the “pattern-match the environment” assistant. It collapses to a ninth of the tracker’s accuracy precisely where builds stop matching templates, reproducing in-domain and under perception

Where this pipeline is stronger	Where game systems are stronger
Perception-in premise: no oracle scene graph, no ground-truth action labels — the regime robots actually face	Absolute goal accuracy: with oracle symbolic state their inference is near-ceiling; MICA pays for perception (0.375 on never-seen sessions)
Explicit, inspectable, <i>swappable</i> belief — the arm swap turns the belief itself into the measured variable	Belief-space planning: POMDP-style systems act to reduce their own uncertainty; MICA only filters and gates, never plans
Calibration measured (reliability, pooled ECE) — a quantity the lineage never reported	Open goal spaces: predicate- or reward-valued goals vs. MICA’s fixed five-category taxonomy
Causally clean traces: do-operator, snapshot rule, no belief feedback — intervention-grade data	Task-outcome evaluation: trained assistance policies show speedup / completion gains; MICA has placed zero blocks (pinned) and claims no efficacy
Consent-gated authority: safety does not depend on the belief being right	Demonstrated early recognition — typically in domains engineered for goal separability [18]; MICA’s earliness is honestly unproven
Label-free improvement paths: real-session pre-training, contested labels under the builder’s word — grows from disputed and unlabeled data	Simulator oracles are cheap; MICA’s proof-grade capture rig is real instrumentation work

Table 1: An honest two-way comparison with intention-recognition game systems. The columns are complementary rather than competing: MICA trades absolute inference accuracy and planning depth for deployability of premise, auditability, and safety that does not presume a correct belief.

what the oracle-state literature reported symbolically [2, 3]. This is RQ2’s necessity half: whatever understanding is, structure lookup alone is not it.

4.2 Claim 2 — The agent anticipates next movements with minimal context

Data. On 1,158 held-out rows from four real sessions fixed by id before any run, the action decoder’s τ -token accuracy (exact for symbolic tokens, ± 2 for coordinates; chance level 0.012 over the 87-token vocabulary): **0.710** one action ahead, 0.621 two ahead, 0.586 four ahead, **0.593 eight ahead**. On the scripted holdout, the multi-token retrofit passed its pre-registered adjudication (stage-B NLL $1.637 < 1.660$ stage-A; four-ahead accuracy 0.743).

Reading. “Minimal context” is literal: frozen encoders, no instruction, no goal specification, no labels at deployment — the decoder sees only the evidence stream and proposes how the build continues. Predicting a free human’s actions eight steps out at fifty times chance is the strongest single number in this thesis, and it is the operational meaning of “understands what the player is doing.”

4.3 Claim 3 — Structural understanding accumulates with observation

Data. As the embodied agent’s first-person raycast scan covered a real build (session 052914), the shape embedding computed from *only the scanned cells* converged monotonically to the full structure’s embedding: cosine 0.12 at 9% coverage $\rightarrow$ **0.92 at 91% coverage** (Fig. 3). Independently, the structure matcher scores 1.0 on the scripted corpus (labels known by construction

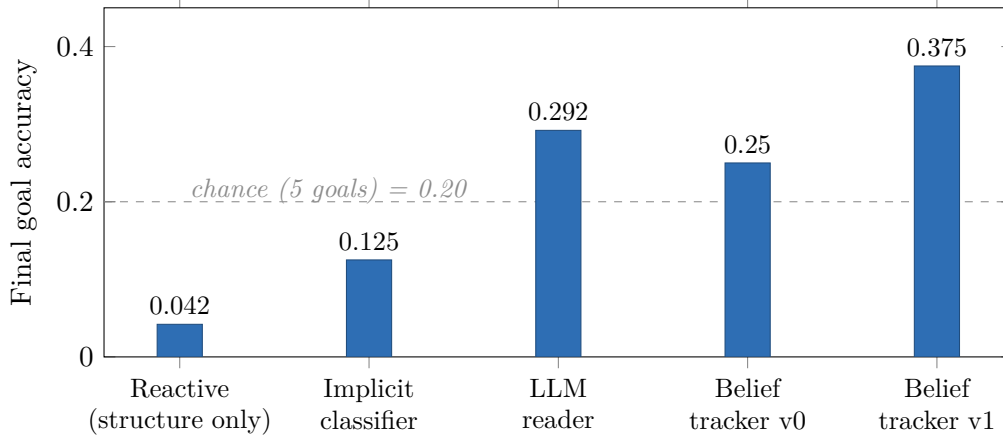


Figure 2: Final goal accuracy on 24 never-seen real sessions. The reactive structure-only baseline collapses to 0.042; the belief tracker (v1) reaches 0.375. (Four-arm comparison, cascade of 2026-07-19.)

— measured, not assumed), and its free-build boundary is mapped rather than hidden: of 37 labeled real sessions, 10 matcher-agreed, 18 contested, 9 discarded.

Reading. This is RQ1’s positive half with its limits attached: where ground truth exists the matcher is exact; where builds go off-template it fails, honestly and measurably — and that boundary is precisely where Claim 1 shows the belief must take over.

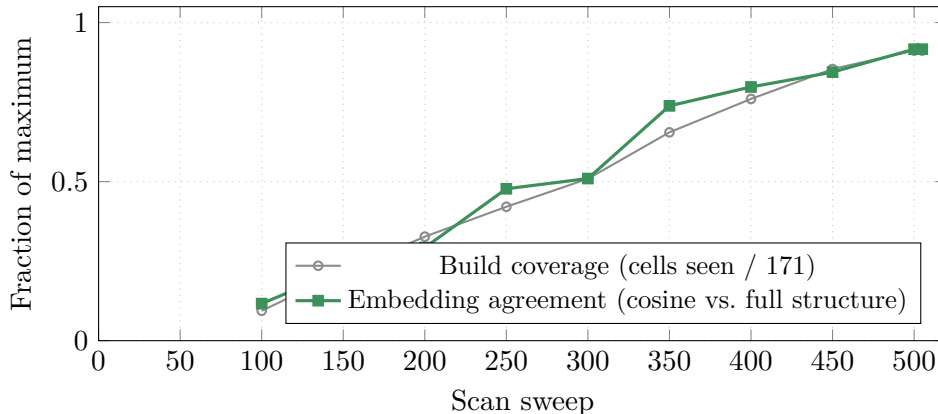


Figure 3: The agent’s partial view carries the full structure’s signal: the embedding of the scanned portion converges to the exact embedding as coverage grows.

4.4 Claim 4 — The system improves lawfully with the right data

Two experiments, both pre-registered with frozen decision rules before any run, both run inside the same deliberate retrain, both won — and neither shipped anything without a separate dated adoption decision.

Data (contested labels). Training the likelihood heads on matcher-contested sessions with the *builder’s* label as truth, against the agreed-only control, on a shared evaluation set neither

arm trained on: clean accuracy $0.000 \rightarrow 0.333$; all-session accuracy $0.182 \rightarrow 0.273$; pooled ECE $0.258 \rightarrow 0.232$; sustained-from (build fraction from which the top goal stays correct; lower = earlier) $0.990 \rightarrow 0.897$. All four pre-registered criteria held. An identical experiment on 2026-07-12 found the *opposite*; the reversal followed the repair of a feature leak and the evidence corpus (§5). Caveat stated with the win: the evaluation set is 11 sessions.

Data (label-free pretraining). The decoder trained only on scripted builds was **1.25 nats worse** on real human action streams than scripted ones (NLL 3.42 vs. 1.77). Because the goal rationale is a separate head outside the token stream, real sessions — contested and discarded included — can pretrain the next-token stage with zero label-noise risk: real-holdout NLL fell to **2.17** (−37%) while the scripted holdout also improved ($1.768 \rightarrow 1.725$). Real streams were the missing distribution, not noise.

4.5 A case study, labeled as such

One live session (the crop farm, Fig. 5 in the figures supplement) shows the mechanism at work: the belief starts at chance because the live pipeline missed the first 46 placements, recovers the correct goal from the remaining evidence, dips honestly when the builder wanders mid-session, and re-locks at $P = 0.84$ as the farm takes shape. Illustrative only; quantitative claims rest on §4.1.

5 What failed, and what it taught

This section carries the other four claims’ credibility.

Early recognition is unproven. The pre-registered criterion — sustained correct belief from $\leq 40\%$ build progress, significantly above chance by exact binomial — fails for *every* arm ($p = 0.058$ –1.0). The system recognizes goals mid-build, not early. This is the primary motivation for more sessions, and goal-recognition-design theory [18] says earliness is partly an environment property — free builds may simply not separate goals early.

Confidence is not calibrated. Above 0.4 stated confidence, held-out reliability *inverts* (the 0.8–0.9 bin hit 6.9% accuracy) — Guo et al.’s overconfidence warning [19] coming true outside their i.i.d. scope. Consequence, taken seriously: the placement threshold sits above the maximum observed confidence, so confidence-licensed placement is honestly unreachable until a recalibration stage passes its own pre-registered bar.

The belief is not prediction fuel. A post-hoc probe (labeled as post-hoc: it was not pre-registered) forced the decoder’s intent slot to the real tracker belief versus an all-zero slot on the real holdout, at horizons 1/2/4/8, on both decoders: **no gain at any horizon** ($|\Delta| \leq 0.001$). The belief is computed from the same evidence the decoder reads, so for forecasting it is conditionally redundant. Its demonstrated value is situation classification (Claim 1) and action licensing — the judgment layer, not the anticipation layer.

A published verdict was reversed — by finding the bug, not by re-arguing. The 2026-07-12 contested-labels experiment concluded the opposite of §4.4. Between the two runs, an idle-state feature was found leaking the scoring target into the heads’ inputs, and the evidence corpus was found mis-generated (19 of 25 matcher verdicts changed on regeneration). The first verdict was honest for its inputs; its inputs were wrong. An interim results table built on the leaked features was retracted outright.

The agent has never assisted. Across every live session, the agent placed *zero* blocks (pinned): all authority stayed at suggestion, and the one acceptance datum collected was *negative*

(lift -0.002 — the unvoiced-proposal control group catching that the builder did the proposed thing without being told). Assistance in this thesis is designed, instrumented, and safety-gated — never evaluated. Given that a confidently wrong helper is worse than none [2], an agent that declines to act without license is the safety design working, not a shortfall.

6 Limitations and scope

Single builder: all real data comes from one person, so there is no inter-annotator reliability and no cross-player generalization claim. Ground truth is the builder’s own label (defended by design: the matcher’s free-build weakness is documented, and the truth hierarchy — builder’s word outranks the matcher — is pinned). No belief-space planning, no environment redesign, no trained assistance policy: the system only filters, gates, and asks. The VLM cross-check in the labeling pipeline is diagnostic only; its 94% validation number is binary clip agreement with a single annotator and licenses nothing about five-way judging.

7 Future work

(1) *Calibration first*: per-bin recalibration on held-out sessions, with the pre-registered bar of monotone reliability and pooled $ECE \leq 0.15$ — the gate to everything else. (2) *Assistance outcomes*: consent-route sessions with real placements, the acceptance signal at meaningful n , and **placement survival** (does an agent-placed block remain in the finished build?) as the outcome measure — already measurable for free from the capture’s actor-attributed block events. (3) *Earliness*: more sessions for statistical power, plus the pre-registered hotbar/selection-moment features (the builder’s material choices as pre-placement evidence). (4) *Transfer*: any domain with discrete observable actions, goals legible from partial structure, and cheap dense workspace observation.

8 Conclusion

An agent can learn to recognize what a person is building and anticipate their next actions from raw observation alone — and the intent-inference layer is what keeps that understanding alive when work stops matching templates. The equal contribution is the apparatus: pre-registration that constrained its author, falsification criteria that fired, a leak found and a verdict reversed in public, and every negative banked beside the positives. That is what it takes for “the model understands” to be a measurement rather than a slogan.

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